PHY101 (Physics)

Recent Papers with their answers

Q1. A photon of green light has a wavelength of 530nm. Find the photos frequency, magnitude of through 1m width.

Answer: To find the frequency of a photon with a wavelength of 530nm, we can use the formula:

c = λv

Where c is the speed of light, λ is the wavelength, and v is the frequency.

Rearranging the formula, we get:

v = c/λ

Plugging in the values, we get:

v = (3 x 10^8 m/s)/(530 x 10^-9 m)

v = 5.66 x 10^14 Hz

Therefore, the frequency of the photon is 5.66 x 10^14 Hz.

To find the magnitude of the photon's energy through a 1m width, we need to know the photon's energy. The energy of a photon can be calculated using the formula:

E = hf

Where E is the energy of the photon, h is Planck's constant, and f is the frequency of the photon.

Plugging in the values, we get:

E = (6.626 x 10^-34 J s) x (5.66 x 10^14 Hz)

E = 3.75 x 10^-19 J

To find the magnitude of the photon's energy through a 1m width, we need to know the photon flux density. The photon flux density is the number of photons passing through a unit area in a unit time. We can calculate the photon flux density using the formula:

Photon flux density = Power / (Area x c)

Where Power is the energy per second, Area is the cross-sectional area of the beam, and c is the speed of light.

Plugging in the values, we get:

Photon flux density = (3.75 x 10^-19 J/s) / (π x (0.5 x 1)^2 m^2 x (3 x 10^8 m/s))

Photon flux density = 1.50 x 10^9 photons/s/m^2

Therefore, the magnitude of the photon's energy through a 1m width is 1.50 x 10^9 photons/s/m^2.

Q3. Find the difference between two waves which interfere constructively and destructively.

Answer: When two waves interfere constructively, the amplitude of the resulting wave is increased. The difference between the two waves is such that their peaks and troughs line up, resulting in an increase in amplitude.

When two waves interfere destructively, the amplitude of the resulting wave is decreased. The difference between the two waves is such that their peaks and troughs are out of phase, resulting in a cancellation of the amplitude.

Q4. Similarity and difference of Galvano meter and electric motor.

Answer: The Galvanometer and Electric Motor are two different devices that have some similarities and differences.

Similarities:

Both devices have a coil of wire and a magnet that interact with each other to produce motion.

Both devices are based on the principle of electromagnetism.

Both devices can be used to measure electrical currents.

Differences:

The Galvanometer is a measuring instrument that is used to detect small currents, while the Electric Motor is a device that converts electrical energy into mechanical energy to produce motion.

The Galvanometer typically has a very low resistance and is used in circuits where a small current needs to be detected, while the Electric Motor has a higher resistance and is used in circuits where electrical energy needs to be converted into mechanical energy.

The Galvanometer typically has a sensitive pointer or meter to indicate the current, while the Electric Motor typically has a rotating shaft that produces mechanical work.

Q5. Which law used in blackbody radiation?

The law used in blackbody radiation is known as Planck's law. This law describes the spectral energy density of electromagnetic radiation emitted by a blackbody at a given temperature. It states that the energy density of radiation emitted by a blackbody at a given wavelength is proportional to the frequency of the radiation and the temperature of the blackbody. Planck's law was first introduced by Max Planck in 1900 as a theoretical explanation for the observed spectral distribution of radiation emitted by heated objects, such as a blackbody. It played a crucial role in the development of quantum mechanics and was later extended to include other phenomena, such as the photoelectric effect.

Q6. Difference between magnetic flux and magnetic flux density, when flux is max and when it is zero?

Magnetic flux and magnetic flux density are related but distinct concepts in magnetism.

Magnetic flux refers to the total magnetic field passing through a given area. It is represented by the symbol Φ and is measured in units of Weber (Wb). Magnetic flux is a scalar quantity and is calculated by multiplying the magnetic field

strength (B) by the area (A) through which it passes and the cosine of the angle between the field and the normal to the surface.

Magnetic flux density, on the other hand, refers to the strength of the magnetic field in a given region. It is represented by the symbol B and is measured in units of Tesla (T). Magnetic flux density is a vector quantity and is calculated by dividing the magnetic flux by the area through which it passes and the cosine of the angle between the field and the normal to the surface.

When magnetic flux is at its maximum, the magnetic field passing through a given area is at its highest level. This could occur when a magnetic field is applied to a material, creating a strong magnetic field that is passing through a surface. When magnetic flux is zero, there is no magnetic field passing through the given area. This could occur when a magnetic field is removed from a material or when a magnetic field cancels out another magnetic field passing through the same area, resulting in no net magnetic field.

Q7. How isotopes change atomic mass?

Isotopes are different forms of the same element that have the same number of protons but a different number of neutrons in their atomic nucleus. Since the number of protons determines the atomic number of an element, isotopes of the same element have the same atomic number. However, the different number of neutrons means that isotopes have different atomic masses.

When calculating the atomic mass of an element, the atomic masses of all the isotopes of that element are taken into account. The atomic mass of an element is the weighted average of the atomic masses of its isotopes, where the weighting is based on the abundance of each isotope in nature.

For example, the element carbon has two naturally occurring isotopes, carbon-12 and carbon-13. Carbon-12 has an atomic mass of 12 amu (atomic mass units) and makes up about 98.9% of all carbon atoms, while carbon-13 has an atomic mass of 13 amu and makes up about 1.1% of all carbon atoms. The atomic mass of carbon is therefore calculated as:

(0.989 x 12 amu) + (0.011 x 13 amu) = 12.01 amu

As can be seen in this example, the presence of isotopes with different atomic masses can cause the calculated atomic mass of an element to differ slightly from the atomic mass of its most abundant isotope. This effect can be more pronounced for elements with a wider range of isotopes or with isotopes that differ significantly in their relative abundance.

Q8. What is magnetic spin moment?

Magnetic spin moment, also known as spin magnetic moment or simply spin moment, refers to the intrinsic magnetic moment associated with the spin of an elementary particle, such as an electron or a proton. This magnetic moment arises from the spin of the particle, which can be thought of as a rotation of the particle about its own axis.

The spin moment of a particle can interact with an external magnetic field, causing the particle to experience a torque that tends to align its spin axis with the direction of the field. This is the basis of many magnetic phenomena, such as ferromagnetism and Para magnetism.

The spin moment of an electron is particularly important in magnetism, as it is the source of the magnetic moment in many materials. In ferromagnetic materials, the spin moments of electrons in neighboring atoms align with each other, producing a net magnetic moment that gives rise to a macroscopic magnetic field. In paramagnetic materials, the spin moments of electrons are randomly oriented, but can be aligned in the presence of an external magnetic field.

The spin moment of a particle is often represented by the symbol μs, and is typically measured in units of Bohr magnetons (μB), where one Bohr magneton is equal to the magnetic moment associated with the electron's orbital motion in the hydrogen atom. The magnitude of the spin moment depends on the type of particle and its intrinsic properties, such as its mass and spin.

Q9. Write down the Chemical behavior of elements.

The chemical behavior of elements refers to how they interact with other elements and compounds in chemical reactions. This behavior is largely determined by the electronic configuration of the elements, which dictates their chemical properties.

Elements in the same group of the periodic table tend to exhibit similar chemical behavior due to their similar electronic configurations. For example, the alkali metals (Group 1) are highly reactive with water and oxygen, while the noble gases (Group 18) are generally unreactive due to their full valence electron shells.

Chemical behavior can also be affected by the presence of other elements or compounds. For example, halogens (Group 17) are highly reactive with alkali metals to form ionic compounds, while nonmetals like carbon and nitrogen can form covalent bonds with each other to form molecules like carbon dioxide and nitrogen gas.

The chemical behavior of an element can also vary depending on its oxidation state, or the number of electrons it has gained or lost in a chemical reaction. For example, iron can exhibit both ferrous (Fe2+) and ferric (Fe3+) oxidation states, which can affect its reactivity with other compounds.

Overall, the chemical behavior of elements is a complex and varied field, but is essential to understanding how different elements can interact with each other to form the wide range of compounds and materials that we observe in the world around us.

Q10. For observing interference, why we use thin slit only? why don't we use thick slit?

Interference is a phenomenon that occurs when two or more waves overlap and combine to form a resultant wave. In order to observe interference, it is necessary to use a device that can create two or more coherent sources of waves that will interfere with each other.

One common device used for this purpose is a double-slit apparatus, which consists of two narrow slits separated by a small distance. When light waves pass through these slits, they diffract and interfere with each other, producing an interference pattern on a screen placed behind the slits.

The reason that thin slits are used instead of thick slits is that thin slits produce a more pronounced diffraction effect, which is essential for creating a clear interference pattern. Thick slits, on the other hand, do not produce as much diffraction, which can result in a weaker interference pattern or no interference pattern at all.

In addition, thin slits allow for a greater degree of control over the interference pattern, as the distance between the slits can be easily adjusted to produce different patterns. Thick slits, by contrast, are less flexible in terms of producing different interference patterns.

Overall, thin slits are preferred for observing interference because they produce a stronger diffraction effect and allow for greater control over the interference pattern.

Q11. Write about plum pudding model that (most important)

a) Who gave give

b) What was its basic features.

c) Why was it wrong?

The plum pudding model was a model of the atom proposed by J.J. Thomson in 1904. The model was based on the idea that atoms were made up of a diffuse cloud of positive charge with electrons embedded within it, much like the plums in a plum pudding.

The basic features of the plum pudding model were that atoms consisted of a positively charged sphere of matter with electrons embedded within it. The positive charge was assumed to be distributed uniformly throughout the sphere, with the electrons occupying the spaces between the positive charges.

Thomson proposed this model based on his discovery of the electron in 1897, which showed that atoms were not indivisible, as had previously been thought. Instead, he believed that atoms were made up of smaller subatomic particles, with the negative electrons held within a positively charged matrix.

However, the plum pudding model was ultimately found to be incorrect. In 1911, Ernest Rutherford and his colleagues performed the famous gold foil experiment, in which they fired alpha particles at a thin sheet of gold foil. They expected the alpha particles to pass through the foil with only minor deflections, in accordance with the plum pudding model. However, they observed that a small fraction of the alpha particles was deflected at large angles, and some even bounced back in the direction they had come from.

This result could not be explained by the plum pudding model, as it predicted that the positive charge was distributed uniformly throughout the atom. Rutherford realized that the positive charge must be concentrated in a small, dense nucleus at the center of the atom, with the electrons orbiting around it. This led to the development of the nuclear model of the atom, which is still the basis of our understanding of atomic structure today.

Q12. composition of sun.

The sun is a massive, luminous ball of gas that is at the center of our solar system. It is composed mostly of hydrogen and helium, with trace amounts of other elements.

About 74% of the sun's mass is hydrogen, while helium makes up about 24%. The remaining 2% is made up of trace amounts of other elements, such as oxygen, carbon, neon, and iron.

The sun's core, where nuclear fusion takes place, is the hottest part of the sun, with temperatures of up to 15 million degrees Celsius. At these temperatures, hydrogen atoms are fused together to form helium, releasing vast amounts of energy in the process. This energy radiates outwards through the sun's layers, eventually reaching the surface and escaping into space as light and heat.

In addition to the elements that make up the sun, it also contains various layers with different characteristics. The outermost layer, known as the corona, is composed of highly ionized gas that extends far into space. Beneath the corona is the chromosphere, which is a layer of cooler, dense gas that is visible during solar eclipses. The layer below the chromosphere is the photosphere, which is the visible surface of the sun and the layer where most of the sun's light and heat is emitted.

Overall, the sun is a complex and dynamic object, and its composition and structure play a crucial role in shaping the behavior of the solar system as a whole.

Mix Shorts

Question. 1: Do you think clock speed is slower than stationery time.

Question. 2: wave speed is function of frequency are you familiar with this, if we increase frequency than what happened to the speed of wave.

Question. 3: Magnetic field exert force or not while in the presence of charge particles?

Question. 4: what is luckawa force and what is range of it.

Question. 5: Numerical tha Neutron, Deuteron electron ki charges maloom krni thi.

Question. 6: Define capacitors resisters etc.

Long Questions:

Question 1. Define Quantum Revolution.

Question 2. Numerical tha magnetic field maloom krni thi.

Question 3. Capacitor parral me connected hen unka formula batane the ur charge maloom krni thi etc.

Question 4. Relativity ka numerical tha planks constant etc.

Question 2: Yes, wave speed is typically proportional to frequency, with higher frequencies corresponding to higher wave speeds. This relationship is expressed by the equation v = fλ, where v is the wave speed, f is the frequency, and λ is the wavelength.

Question 3: Yes, magnetic fields can exert a force on charged particles that are moving through them. This is the basis of many important phenomena, such as the operation of electric motors and the behavior of charged particles in particle accelerators.

Question 6: Capacitors and resistors are both electrical components that are used in electronic circuits. A capacitor stores electrical charge and is characterized by its capacitance, which is measured in farads (F). A resistor limits the flow of electrical current and is characterized by its resistance, which is measured in ohms (Ω).

Long Questions:

Question 1: The quantum revolution refers to a period of scientific discovery in the early 20th century that fundamentally changed our understanding of the nature of matter and energy. It was characterized by the development of quantum mechanics, a new branch of physics that describes the behavior of particles at the atomic and subatomic level. Key figures in the quantum revolution include Max Planck, Albert Einstein, Niels Bohr, Werner Heisenberg, and Erwin Schrödinger.

Question 3: To calculate the total charge on capacitors connected in parallel, you can use the equation Q = CV, where Q is the total charge, C is the capacitance of each capacitor, and V is the voltage applied to the capacitors. The charge on each capacitor is equal to the product of its capacitance and the voltage applied to it. Resistors, on the other hand, limit the flow of electrical current and are characterized by their resistance, which is measured in ohms (Ω).

MIX

1. Does your bathroom mirror show your older and younger than your actually.

2. What is the advantage of transmitting power at high voltage, will the transformer operate, if a battery is used for input voltage primary.

3. If the field charge inside a closed surface is known both the distribution of the charge is unsatisfied, can you use Gauss law to find electric filed.

4. Many traffic lights charge when a car roll up intersection. How dose light sense the presence of car.

ANSWERS

No, your bathroom mirror does not show you as older or younger than you actually are. However, mirrors do reflect an image that is flipped horizontally, so your left side appears on the right and vice versa.

Transmitting power at high voltage has the advantage of reducing energy losses due to resistance in the transmission lines. This is because power loss is proportional to the square of the current, and higher voltages allow for lower currents to achieve the same power transmission. A transformer will only operate if there is a changing magnetic field, so a battery would not be able to drive the primary coil of a transformer.

Yes, Gauss's law can be used to find the electric field inside a closed surface if the charge distribution is known. Gauss's law states that the total electric flux through a closed surface is proportional to the charge enclosed by the surface. By applying Gauss's law, it is possible to find the electric field at any point inside the closed surface.

Many traffic lights use sensors to detect the presence of vehicles at an intersection. These sensors can be inductive loops embedded in the roadway, which detect changes in magnetic fields as vehicles pass over them. They can also be infrared sensors that detect the presence of vehicles by measuring changes in the infrared radiation emitted by the vehicles. Once the presence of a vehicle is detected, the traffic light can change accordingly to allow for safe and efficient traffic flow.

1. What becomes of isotopes that undergo alpha decay? Which produces a higher counting rate or radiation dectr a gram of radioactive having a short half-life r gram having a long half-life.
2. Explain solar zone. What is its other name?
3. Distinguish between electric potential and electric potential energy. A negative charge move in the direction of uniform.
4. What is Yukawa Force and its range
5. Define thermometer, internal energy, Heat
6. What is the pressure of sun?

Answers:

1. Isotopes that undergo alpha decay emit alpha particles, which are helium nuclei consisting of two protons and two neutrons. As a result, the atomic number of the parent nucleus decreases by two, while the atomic mass decreases by four. The daughter nucleus produced by alpha decay has a lower energy state than the parent nucleus. A gram of radioactive material with a short half-life will produce a higher counting rate of radiation compared to a gram of radioactive material with a long half-life because it undergoes radioactive decay more rapidly.

2. The solar zone is the region of the sun's interior where energy is transported by radiation. This region extends from the center of the sun to about 0.7 solar radii. It is also known as the radiative zone because the energy is transported by radiation rather than convection.

3. Electric potential is the electric potential energy per unit charge at a point in an electric field. It is a scalar quantity and is measured in volts. Electric potential energy, on the other hand, is the energy that a charged particle possesses due to its position in an electric field. It is a scalar quantity and is measured in joules. If a negative charge moves in the direction of a uniform electric field, it will experience a force in the opposite direction to the field, and its potential energy will decrease.

4. The Yukawa force, also known as the nuclear force, is a strong attractive force between nucleons (protons and neutrons) that holds the atomic nucleus together. It is mediated by the exchange of mesons, which are subatomic particles with integer spin. The range of the Yukawa force is very short, on the order of 10^-15 meters, because the exchange particles have a relatively large mass.

5. A thermometer is a device used to measure temperature. Internal energy is the total energy of a system due to the motion and interactions of its particles. Heat is the energy that is transferred between two bodies at different temperatures when they are brought into thermal contact with each other.

6. The pressure at the center of the sun is estimated to be about 250 billion atmospheres. However, the pressure decreases rapidly as you move outwards from the center, reaching about 1 atmosphere at the visible surface of the sun, known as the photosphere.